

## Photon Theory

- Intensity (I) =  $\frac{\text{Power}}{\text{Area}} = \frac{E}{tA}$   
= energy per unit area per unit time
- Point source  $I = \frac{P}{4\pi r^2} \Rightarrow I \propto \frac{1}{r^2}$
- Line source  $I = \frac{P}{2\pi r l} \rightarrow I \propto \frac{1}{r}$

- no. of Photons  
 $E = n h\nu \Rightarrow n = \frac{E}{h\nu} = \frac{E\lambda}{hc}$
- no. of Photons per unit time  
 $\frac{n}{t} = \frac{E}{th\nu} = \frac{P}{h\nu} = \frac{IA}{h\nu} = \frac{IA\lambda}{hc}$
- no. of Photons per area per unit time  
 $\frac{n}{At} = \frac{E}{tAh\nu} = \frac{P}{Ah\nu} = \frac{I}{h\nu} = \frac{I\lambda}{hc}$

- Power of incident radiation (P)

$$P = \frac{n h c}{\lambda} \quad P \propto \left(\frac{n}{\lambda}\right)$$

source 1  $P_1 \rightarrow \lambda_1 \rightarrow n_1$   
source 2  $P_2 \rightarrow \lambda_2 \rightarrow n_2$

$$\frac{P_1}{P_2} = \frac{n_1}{n_2} \times \frac{\lambda_2}{\lambda_1}$$

## Dual Nature of Radiation

- Momentum of photon (p) =  $\frac{E}{c} = \frac{h}{\lambda}$
- Force (F) =  $\frac{\Delta p}{\Delta t}$
- Radiation Pressure =  $\frac{F}{A}$
- For perfectly reflecting surface  
 $\Delta p = \frac{2E}{c} \quad F = \frac{2P}{c}$   
Rad. Pressure =  $\frac{2I}{c}$
- For Perfectly Absorbing Surface  
 $\Delta p = \frac{E}{c} \quad F = \frac{P}{c}$   
Rad. Pressure ( $P_R$ ) =  $\frac{I}{c}$

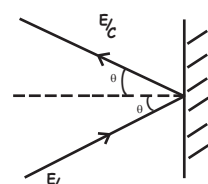
## DUAL NATURE OF RADIATION & MATTER

- Perfectly Reflecting at an angle

$$\Delta p = \frac{2E}{c} \cos \theta$$

$$F = \frac{2P}{c} \cos \theta$$

$$\text{Rad. Pressure} = \frac{2I}{c} \cos \theta$$



### PHOTOELECTRIC EFFECT

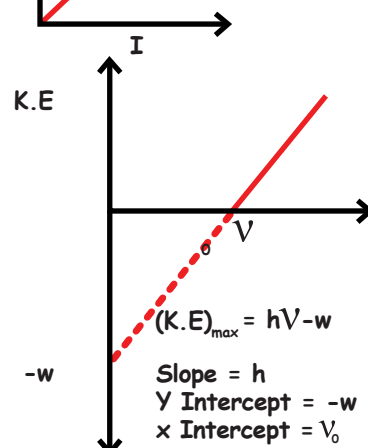
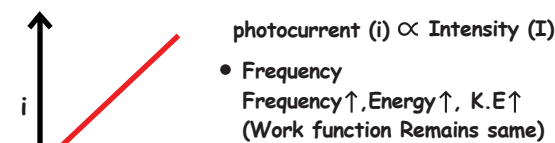
- Energy of photon  $E = h\nu$
- $\nu$  = Frequency of incident light in Hz =  $\frac{c}{\lambda}$
- Max. kinetic energy of emitted photoelectron  
 $(K.E.)_{\max} = E - w = \frac{1}{2} m v_{\max}^2$

### Work function (w)

- Minimum energy required for photoelectric effect to occur  
 $w = h\nu_0 = h \frac{c}{\lambda_0} \quad h = 6.63 \times 10^{-34} \text{ Js}$   
 $\nu_0$  = Threshold frequency in Hz  
 $\lambda_0$  = Threshold wavelength in m
- Work function only depends on nature of metal

### Factors affecting photoelectric effect

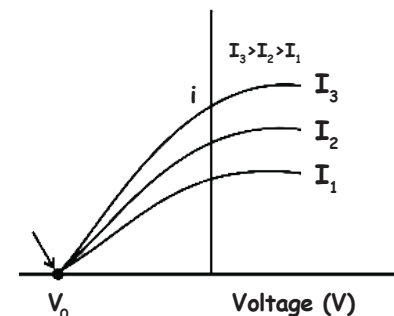
- Intensity  
Intensity  $\uparrow$ , photoelectrons  $\uparrow$ , photocurrent  $\uparrow$  (K.E Remains same)



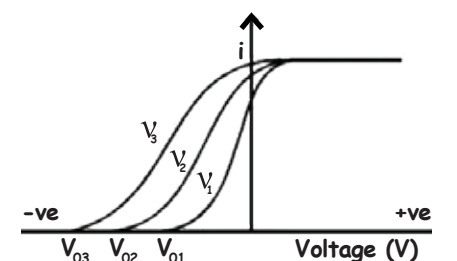
- Anode potential  
Opposes K.E of electron  
Max Negative anode potential = Stopping potential ( $V_0$ ) for which Photocurrent (i) = 0

### Factors affecting stopping potential

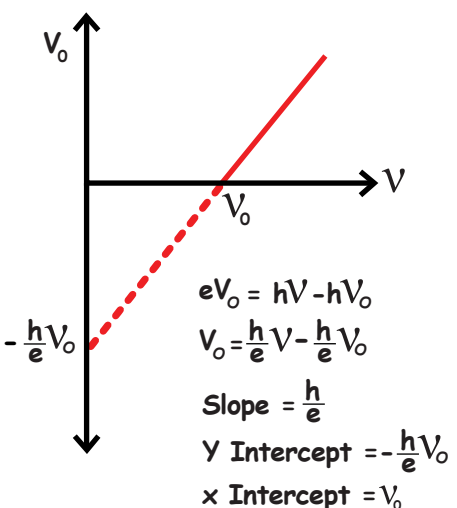
- Intensity (I)  
Intensity  $\uparrow$ , K.E Remains same  
Stopping potential remains same



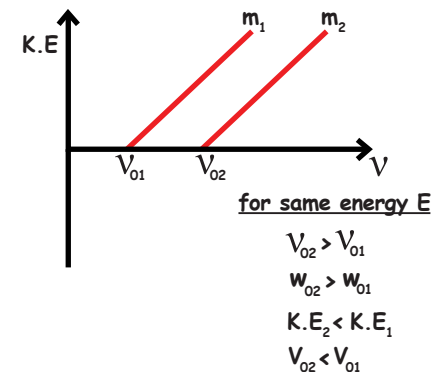
- Frequency  
Frequency  $\uparrow$ , Energy  $\uparrow$ , K.E  $\uparrow$ ,  $V_0 \uparrow$



### Stopping potential $V_0$ vs frequency graph

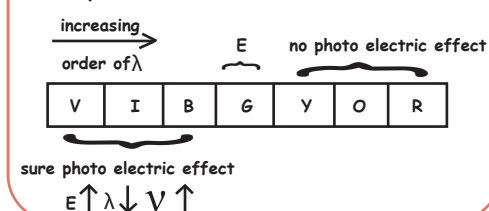


- Nature of material



- Conceptual question

If green color have just sufficient energy for photoelectric effect



### Useful conversions

- Wave Length (nm)  $\rightarrow$  K.E (eV)  
 $K.E = \frac{1240}{\lambda} \text{ (eV)}$
- Wave Length ( $\text{\AA}$ )  $\rightarrow$  K.E (eV)  
 $K.E = \frac{12400}{\lambda} \text{ (eV)}$

- Two Identical photo cathode receive light of frequencies  $f_1$  &  $f_2$ . If velocity of photo electrons are  $v_1$  &  $v_2$  then  $v_1^2 - v_2^2 = \frac{2h}{m} [f_1 - f_2]$

## Dual Nature of Matter

### Wave nature of Matter

Debroglie waves  
fast moving particles like electron with much smaller mass behaves like a wave ie., Circular stationary waves

$$E = mc^2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mc} \rightarrow \lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2m(K.E)}} = \frac{h}{\sqrt{3mk_B T}} = \frac{h}{\sqrt{2mqV}}$$

K.E =  $qV$  (for charged particle)  
 $V$  = accelerating potential in Volt  
K.E =  $\frac{3}{2} K_B T$  (thermal neutron)  
 $K_B$  = Boltzmann's constant  
 $T$  = Temperature in Kelvin

- Thermal Neutron

$$\lambda = \frac{h}{\sqrt{3mk_B T}} = \frac{30.83}{\sqrt{T}} \text{\AA}$$

- Electron  $\lambda = \frac{12.27}{\sqrt{V}} \text{\AA}$

- Proton  $\lambda = \frac{0.286}{\sqrt{V}} \text{\AA}$

- Deuteron ( $\lambda$ ) =  $\frac{0.202}{\sqrt{V}} \text{\AA}$

- $\alpha$ -Particle ( $\lambda$ ) =  $\frac{0.101}{\sqrt{V}} \text{\AA}$

$^1\text{H}$  (Proton)  $\rightarrow$  1 Proton  $\rightarrow m, q$

$^2\text{H}$  (Deuteron)  $\rightarrow$  1 Proton + 1 Neutron  $\rightarrow 2m, q$

$^4\text{He}$  ( $\alpha$ -Particle)  $\rightarrow$  2 Proton + 2 Neutron  $\rightarrow 4m, 2q$

### Relationship b/w wavelength of photon & that of electron

- Ratio of wavelength of photon to that of electron with same energy E

$$\frac{\lambda_{\text{photon}}}{\lambda_e} = c \sqrt{\frac{2m}{E}} \quad \lambda_{\text{photon}} \propto \lambda_e^2$$

- Ratio of K.E of electron to that of photon with same wavelength

$$\text{for Same } \frac{K.E_e}{K.E_{\text{photon}}} = \frac{V}{2C}$$

- A particles formed due to completely inelastic collision of particle 'x' and 'y' having debroglie wave length  $\lambda_x$  and  $\lambda_y$  respectively.

If they are moving in opposite directions

$$p = p_x - p_y \quad \overrightarrow{p_x} \quad \overleftarrow{p_y}$$

then

$$\frac{h}{\lambda} = \frac{h}{\lambda_x} - \frac{h}{\lambda_y} \quad \lambda = \frac{\lambda_x \lambda_y}{\lambda_x - \lambda_y}$$

- If they are moving at right angle to each other

$$\Rightarrow p = \sqrt{p_x^2 + p_y^2} \rightarrow \frac{h}{\lambda} = \sqrt{\frac{h^2}{\lambda_x^2} + \frac{h^2}{\lambda_y^2}}$$

$$\frac{1}{\lambda} = \sqrt{\frac{1}{\lambda_x^2} + \frac{1}{\lambda_y^2}}$$